

MACHINE LEARNING FUNDAMENTALS

MINI PROJECT TOPICS, PROBLEM STATEMENTS & REQUIREMENTS

Student Project Allocation and Development Guidelines

Course	Machine Learning Fundamentals
Group Size	4-5 Students
Number of Groups	15
Project Title	Each group will create its own final project title

Purpose of the Mini Project

Each group will select a Machine Learning problem from the assigned problem domain and develop the solution independently. Students are not required to use the sample titles shown in this document. The sample domain is provided to guide project selection. The group must create its own project title, select a suitable dataset, build the ML model, evaluate it and deploy the final model using Streamlit.

- The final title must be meaningful and directly related to the problem.
- The selected project and dataset should be approved by the faculty before implementation.
- Students must build and understand their own model and code.
- At least one suitable Machine Learning algorithm is compulsory.
- Comparison of two suitable models is recommended where applicable.
- Streamlit deployment is compulsory.
- Every member must participate and be prepared for individual viva.

GROUP 12 - Retail Sales Prediction

Problem Statement: Build a model that predicts retail sales using historical sales, product, store, promotion, month, holiday and other suitable information.

Minimum Requirements:

- Use sales as the target.
- Perform EDA and appropriate date/time feature extraction.
- Perform preprocessing and feature engineering.
- Implement Linear Regression and/or Polynomial Regression.
- Evaluate using MAE, MSE, RMSE and R^2 .
- Deploy the prediction model using Streamlit.

Final Project Title: The group must create its own title based on this problem domain. The sample domain name above is not the compulsory title.

Common Requirements for All Groups

- Problem definition and clear objectives.
- Relevant dataset with source, feature description and target variable.
- Data cleaning and preprocessing appropriate to the dataset.
- Exploratory Data Analysis with meaningful visualizations and interpretations.
- Feature selection or feature engineering where appropriate.
- Training and testing of a suitable Machine Learning model.
- Use of appropriate evaluation metrics.
- Model comparison and final model selection where applicable.
- Saving/loading the trained model correctly.
- Functional Streamlit application with user inputs and output.
- Testing with multiple inputs and basic input validation.
- Technical report, source code, dataset, model file, requirements.txt and README.
- Individual contribution record and individual viva.

Expected Project Development Flow

Problem → Dataset → Data Understanding → Data Cleaning → Preprocessing → EDA → Feature Engineering → Train/Test Split → Model Building → Evaluation → Model Comparison → Best Model → Model Saving → Streamlit Deployment → Testing

Suggested Internal Evaluation - 100 Marks

Component	Marks
Problem Definition & Objectives	10
Dataset & Data Understanding	10
Data Preprocessing	15
EDA & Visualization	10
ML Model Development	20
Evaluation & Model Comparison	15
Streamlit Deployment	10
Report & Documentation	5
Presentation & Viva	5
TOTAL	100

Minimum Report Contents

- Cover Page
- Abstract
- Introduction and Motivation
- Problem Statement and Objectives
- Literature Survey
- Existing System / Limitations
- Proposed System and Workflow
- Dataset Description
- Data Preprocessing
- Exploratory Data Analysis
- Model Development
- Model Evaluation and Comparison
- Streamlit Deployment
- Results and Discussion
- Conclusion and Future Scope

- References

Final Submission Checklist

- Approved self-selected project title
- Problem statement and objectives
- Dataset and source
- Python notebook/source code
- Preprocessing and EDA
- Trained ML model
- Evaluation and comparison results
- Functional Streamlit application
- requirements.txt and README
- Project report PDF
- Application screenshots
- Individual contribution details

Important

The purpose of the mini project is to demonstrate the complete Machine Learning lifecycle. Students will be assessed on their understanding and implementation, not only on model accuracy. Copied projects or copied notebooks will not be considered as original work. Each student should be able to explain the dataset, preprocessing, algorithm, evaluation and Streamlit deployment during the viva.